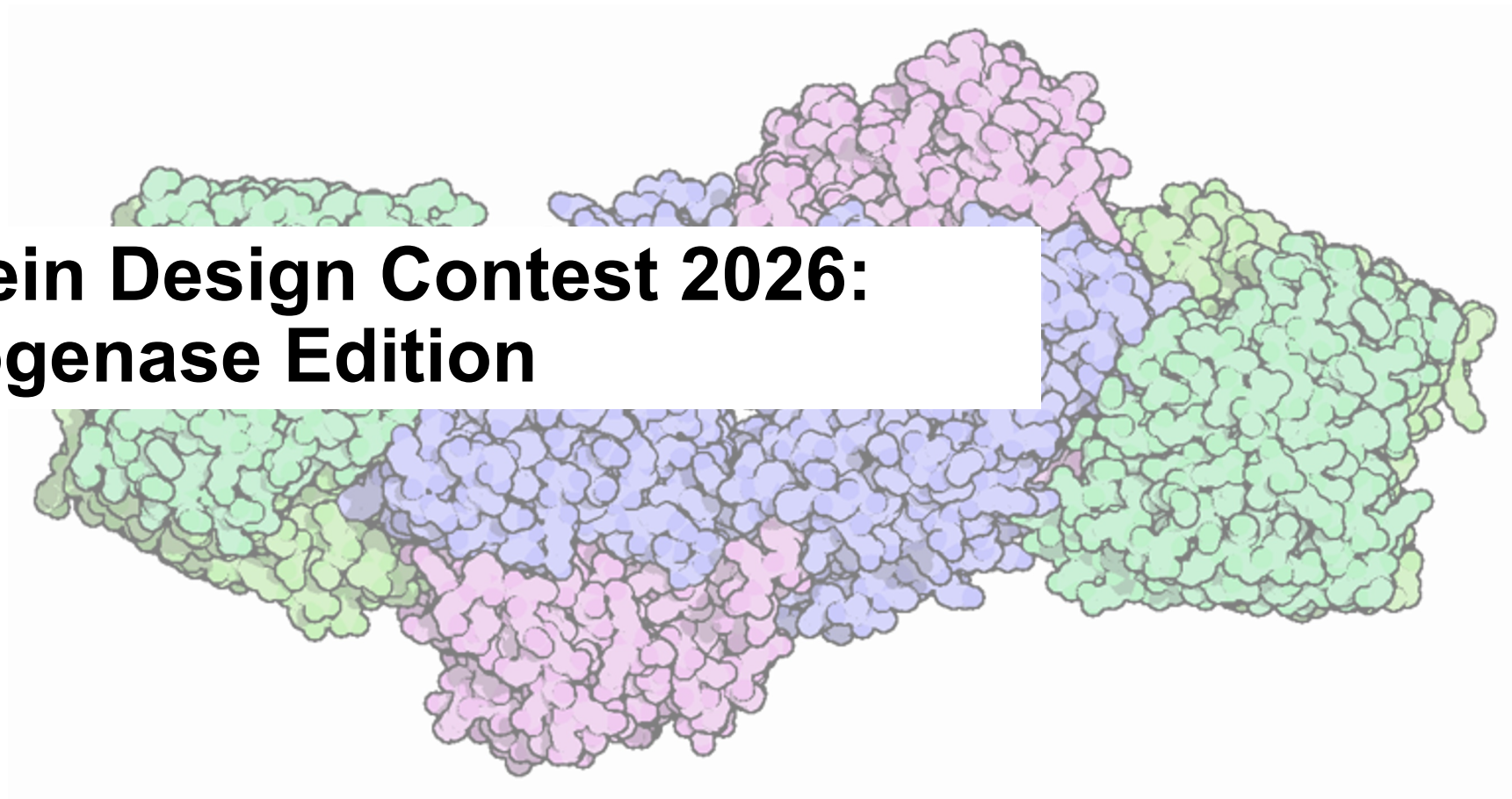
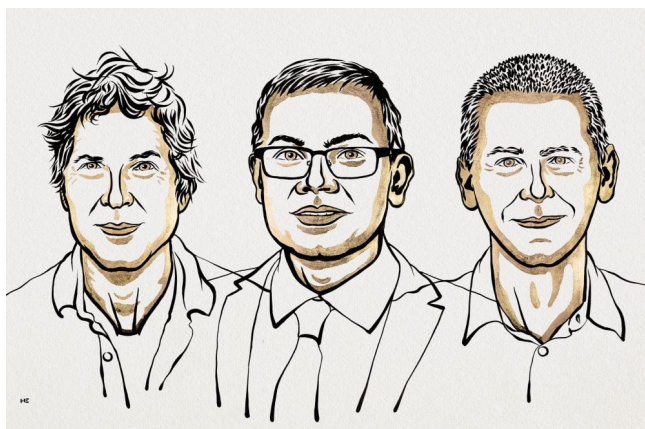


Protein Design Contest 2026: Nitrogenase Edition



2024 Nobel Prize in Chemistry



David Baker

University of Washington, Seattle, WA, USA
Howard Hughes Medical Institute, USA

"for computational protein design"

and the other half jointly to

Demis Hassabis

Google DeepMind, London, UK

John Jumper

Google DeepMind, London, UK

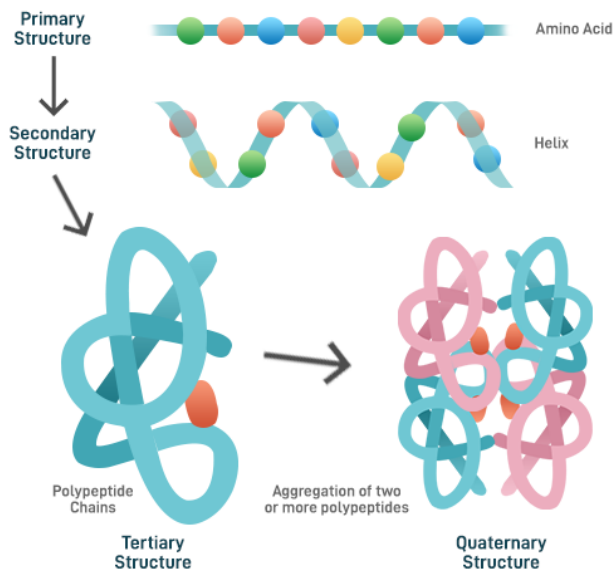
"for protein structure prediction"

They cracked the code for proteins' amazing structures

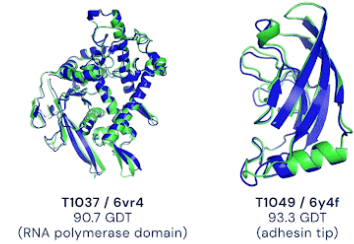
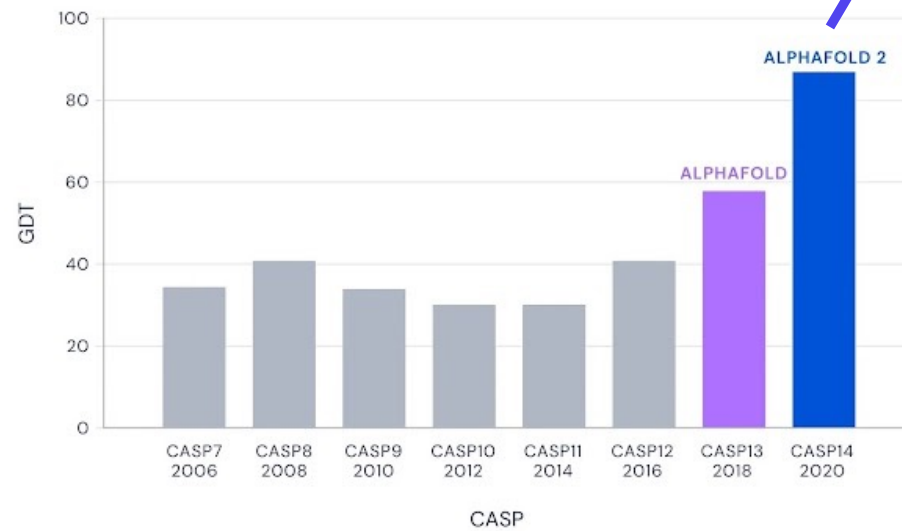
The Nobel Prize in Chemistry 2024 is about proteins, life's ingenious chemical tools. David Baker has succeeded with the almost impossible feat of building entirely new kinds of proteins. Demis Hassabis and John Jumper have developed an AI model to solve a 50-year-old problem: predicting proteins' complex structures. These discoveries hold enormous potential.

Protein structure prediction

PROTEIN STRUCTURE

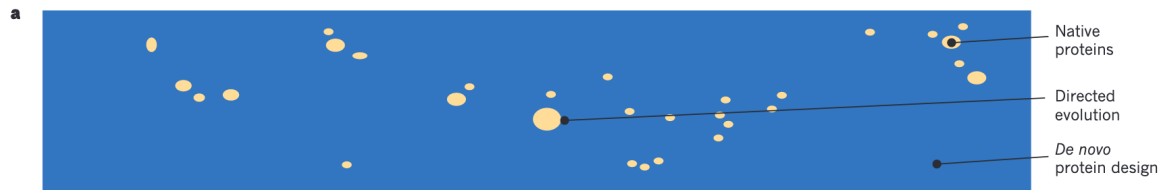


Median Free-Modelling Accuracy

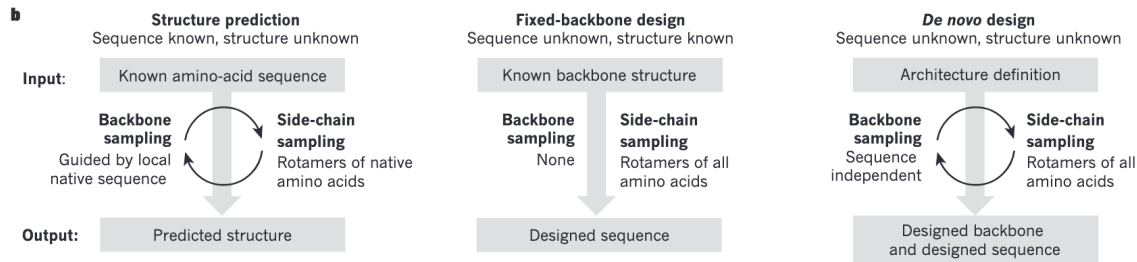


● Experimental result
● Computational prediction

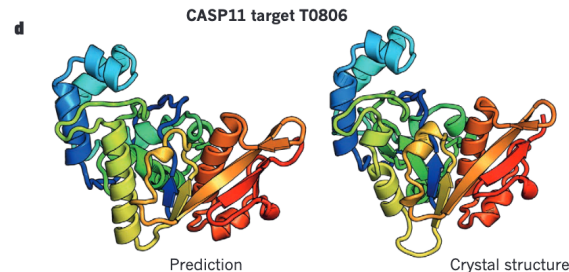
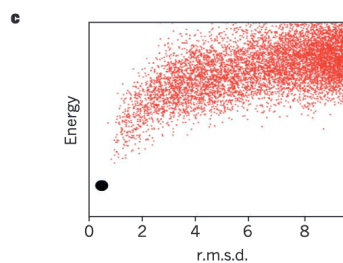
De novo protein design space



Out of a possible 20^{200} sequence space (for 200 aa proteins), only $O(10^{12})$ have naturally occurred.

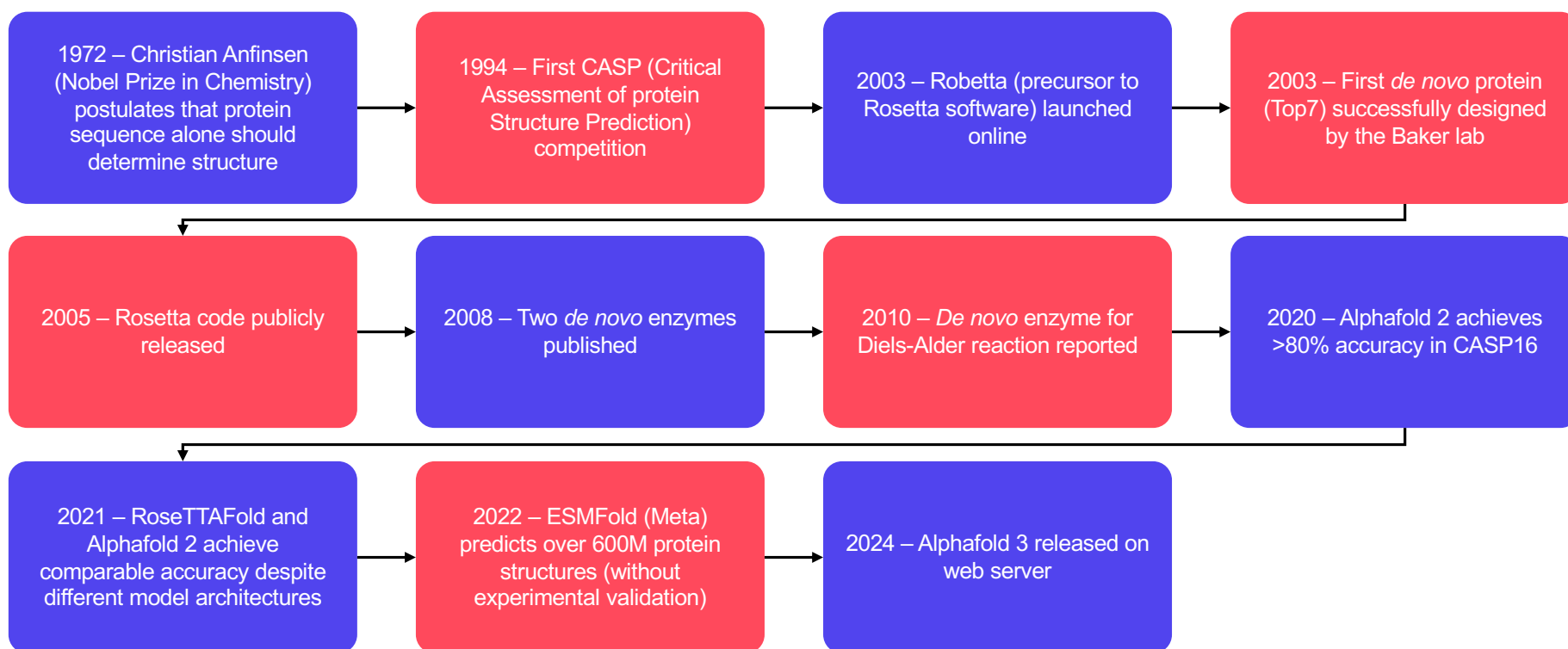


De novo design is a problem with similar degrees of freedom to that of protein structure prediction



Minima in the energy landscape typically corresponds to most likely structures

Evolution of protein structure prediction and design over time

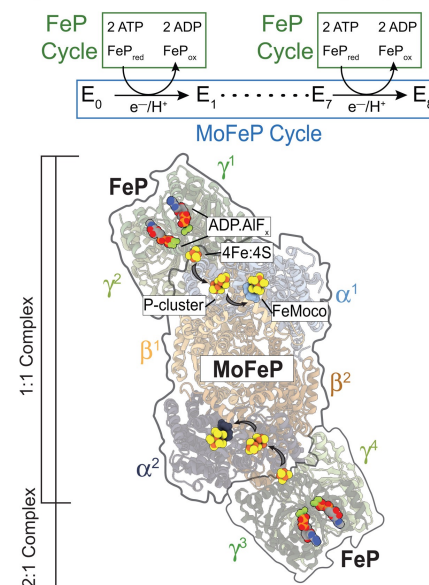


Protein Design Contest 2026

Design Task: Can you create a *de novo* protein that binds with high affinity and specificity to the nitrogen fixing enzyme *nitrogenase* without hindering function?

Rules:

- Maximum sequence length of 100 amino acids
- Individuals or teams up to 4 (mix of UG/Grad okay)
- Submissions should include:
 - Sequence (in FASTA format)
 - Predicted structure (bound to nitrogenase) in PDB format; each protein with a different chain ID – **keep the same chain IDs from 7UT8 and include your protein as chain G**
 - A “pretty” visualization of the protein complex



Nitrogenase Complex

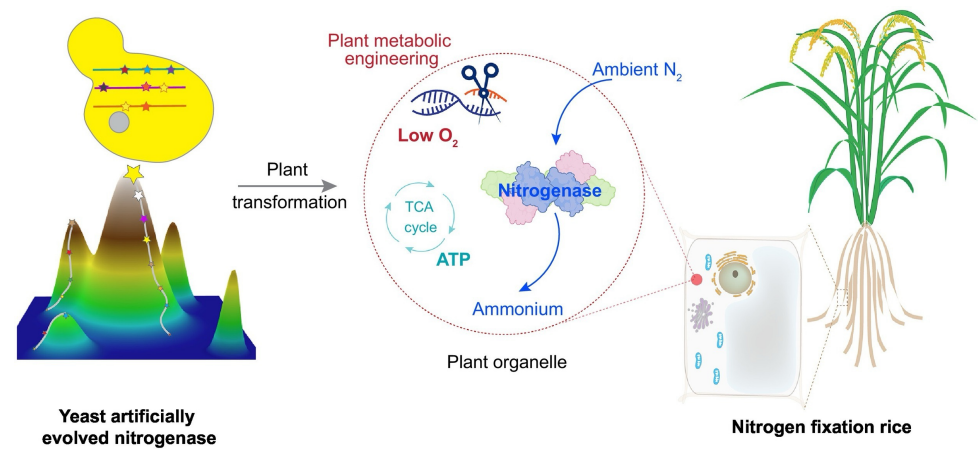
What is nitrogenase?

Nitrogenases are enzymes (in bacteria) that reduce atmospheric N_2 into ammonia (NH_3), a process known as *nitrogen fixation*.

A long-standing “dream”: Haber-Bosch → Biological nitrogen fixation. One specific application is to engineer bacteria that can enhance nitrogen fixation in plants (to increase bioenergy, reduce fertilizer dependence, cut pollution, and ensure global food security).

The challenge? Nitrogenase is inherently oxygen-sensitive (O_2 lowers activity), but plants are in oxygen-rich environments.

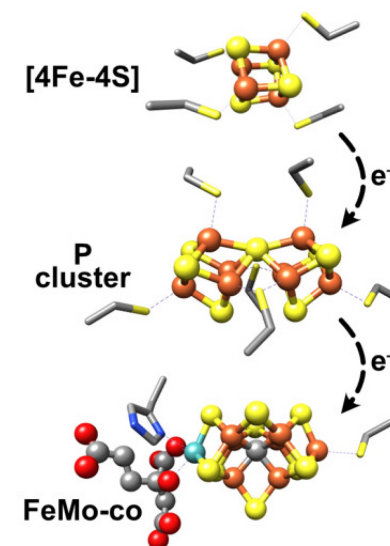
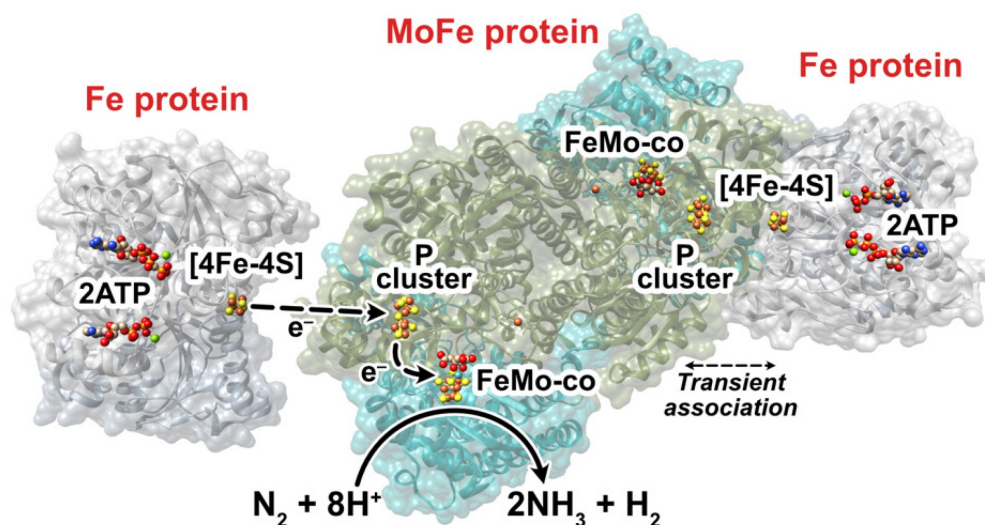
One potential solution? Promote compartmentalization of nitrogenase into low-oxygen environments. Designing protein linkers that can “lasso” nitrogenases via non-covalent interactions is one step forward in this direction!



How does nitrogenase work?

MoFeP is a heterotetramer with 2 α (NifD, cyan) and 2 β (NifK, olive green) subunits.

FeP is a homodimer of NifH (grey).



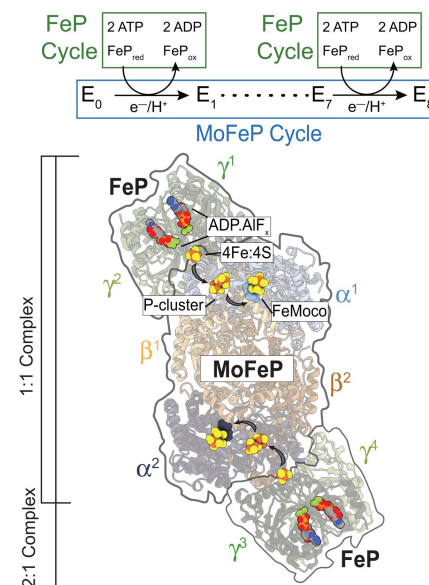
Any designed binder should NOT interfere with FeP binding, MoFeP formation, or the catalytic clusters.
Your best bet is to bind to surface of NifK that does not interact with NifD or NifH

Protein Design Contest 2026

Design Task: Can you create a *de novo* protein that binds with high affinity and specificity to the nitrogen fixing enzyme *nitrogenase* without hindering function?

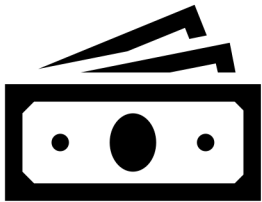
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Nitrogenase Complex

Judging criteria



1st Place
\$150

2nd Place
\$100

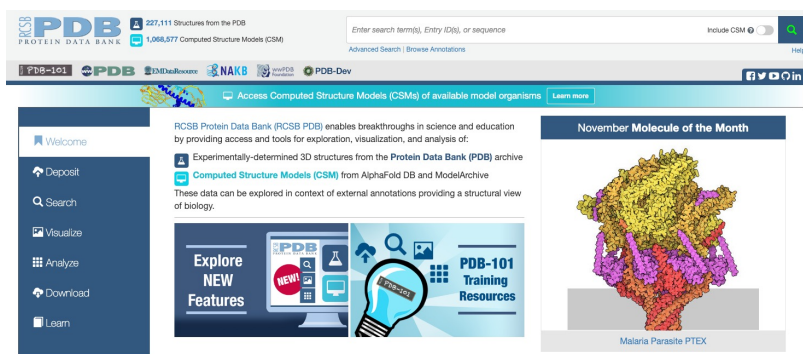
3rd Place
\$50

Rubric:

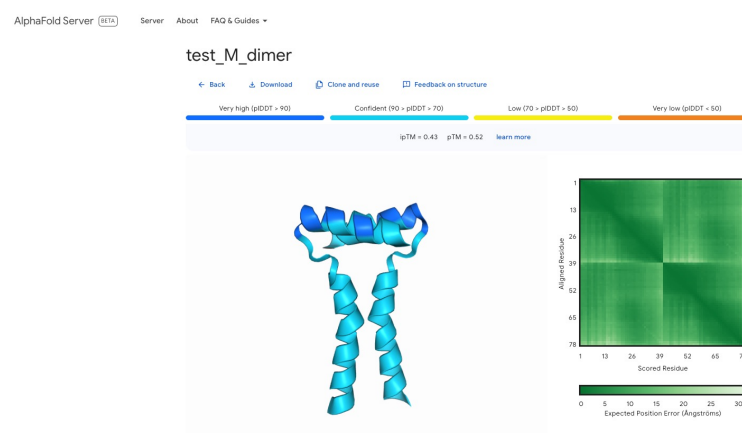
- Interface Quality (50%): How stable is the interaction between the binder and nitrogenase?
- Non-interference (10%): How likely is it that the binder does not interfere with activity?
- Foldability (40%): How likely is the binder to express and fold properly?
- Aesthetics (tie breaker): How visually pleasing is the rendered image of the complex?

(Superlatives are also in play - most creative, most broken, etc. 😊)

Open/accessible tools that you can use

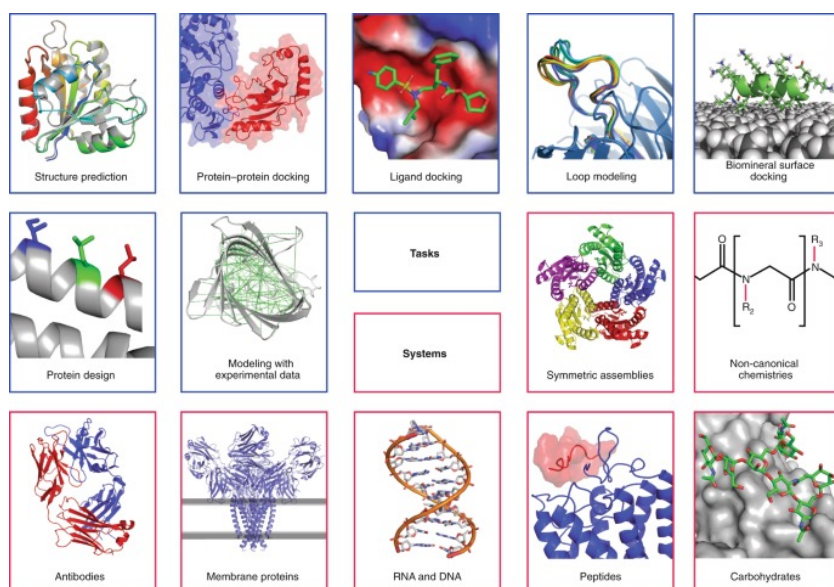


Protein Data Bank – online database containing experimentally-solved structures of proteins via X-ray crystallography, NMR, cryo-EM, etc.

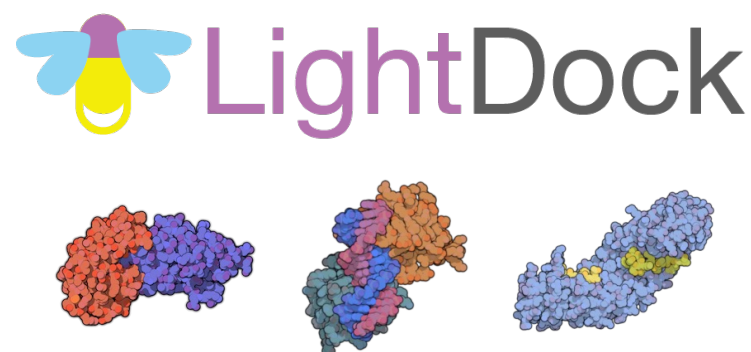


Alphafold 3 – online server to run the AF3 machine learning model from Google DeepMind to predict protein structures from sequences alone.

Open/accessible tools that you can use

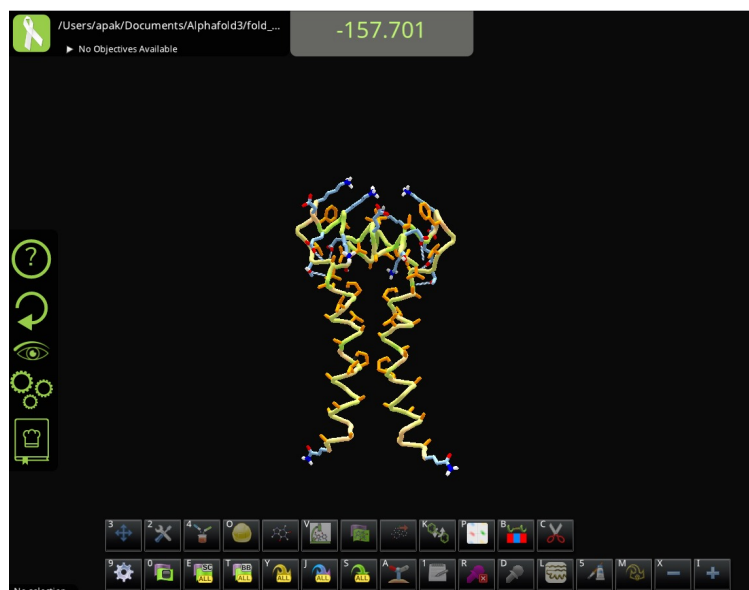


PyRosetta – A suite of Python libraries to perform protein structure modeling, docking, and design.

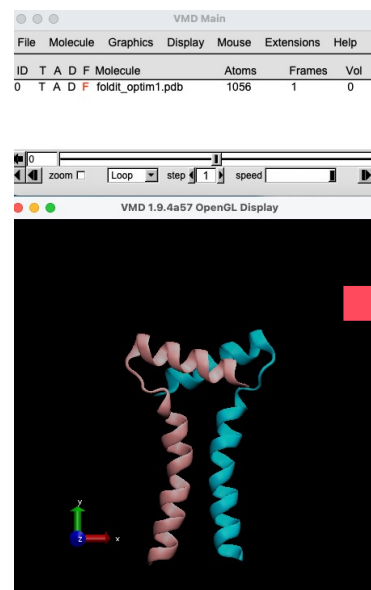


LightDock – An ultrafast, lightweight Python library for protein docking. (There are many other docking software/servers out there)

Open/accessible tools that you can use

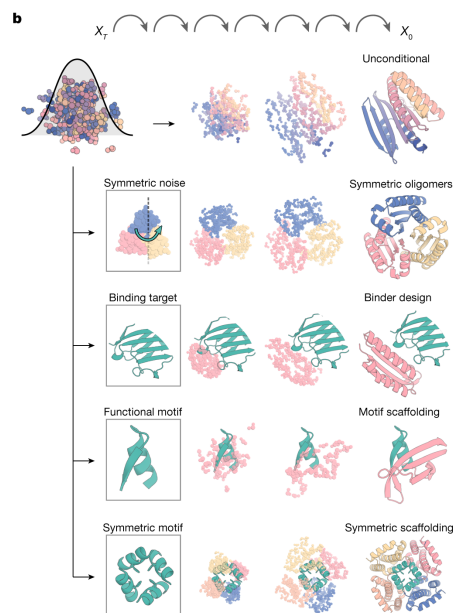


FoldIt Standalone – a GUI tool that offers a nice interface for Rosetta-based modeling (structure optimization/scoring and mutations).

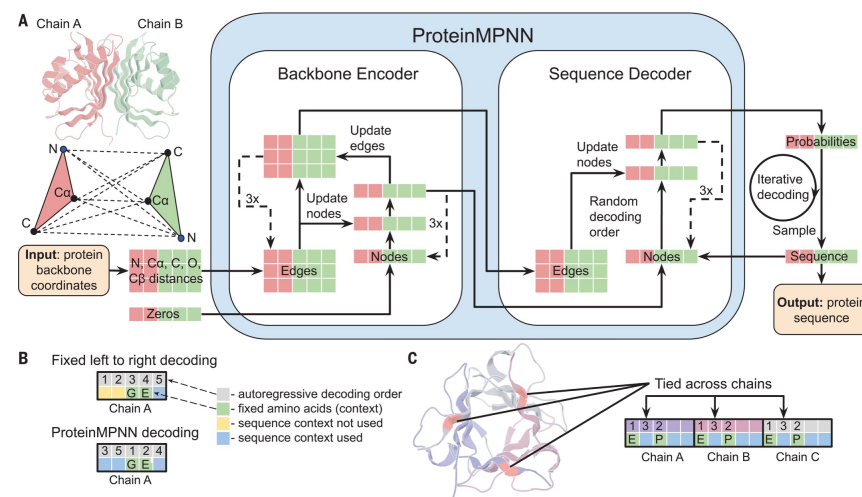


Visual Molecular Dynamics (VMD) – a GUI tool that allows you to visualize, analyze, and render protein structures. Feel free to use similar tools, such as Chimera or pyMOL.

Open/accessible tools that you can use

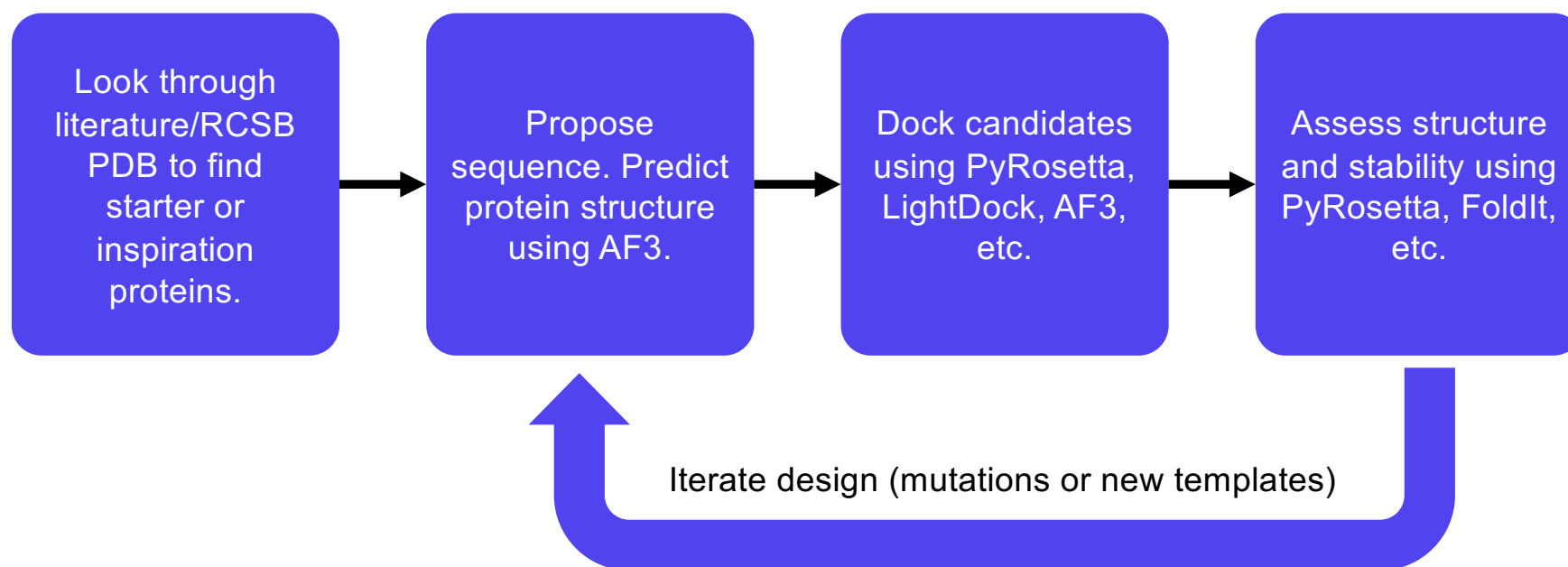


RFdiffusion – A generative ML model to predict protein backbones.

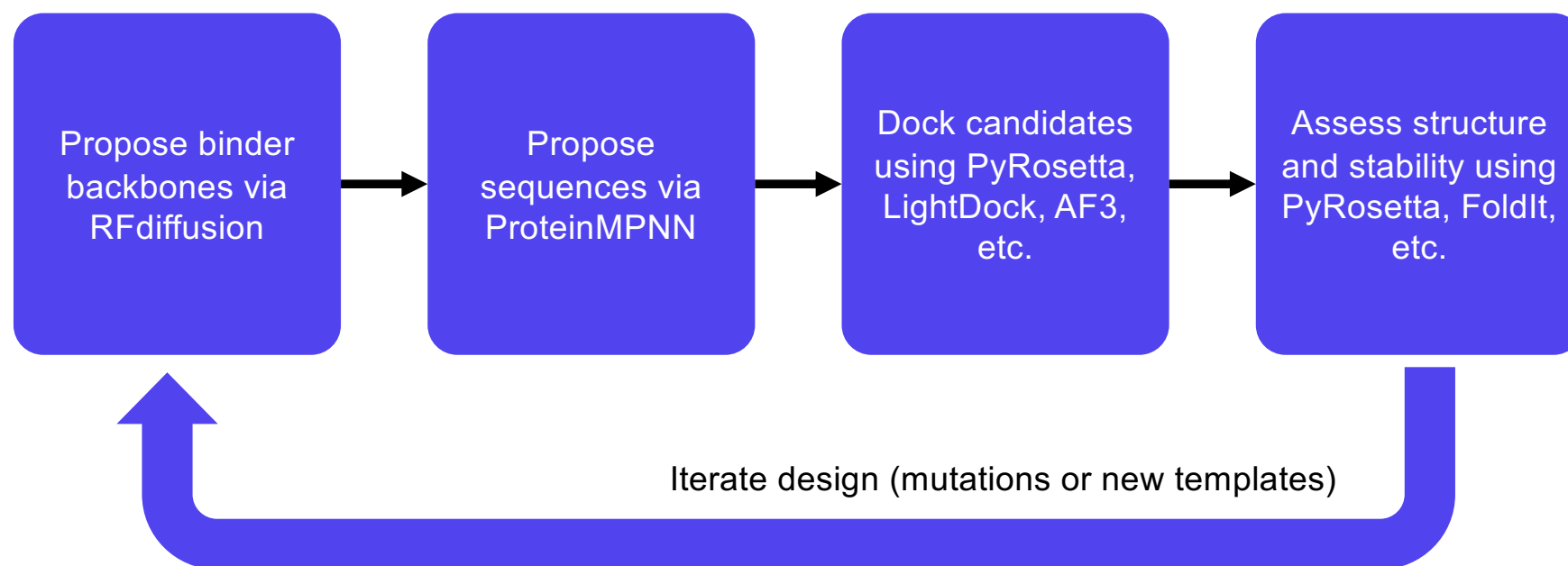


ProteinMPNN – A generative ML model to predict protein sequence given a template (i.e., backbone) structure.

One possible protein design workflow

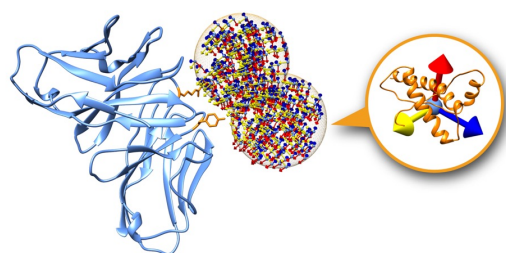


Another possible protein design workflow



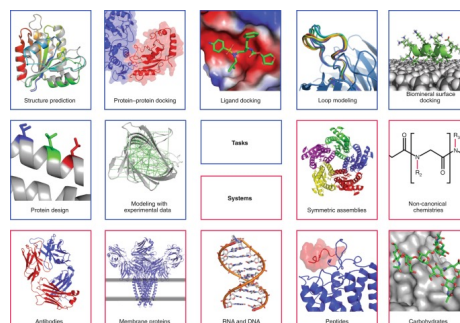
Workshops (all in CTLM 102)

Feb 27 (Fri 4-5pm)



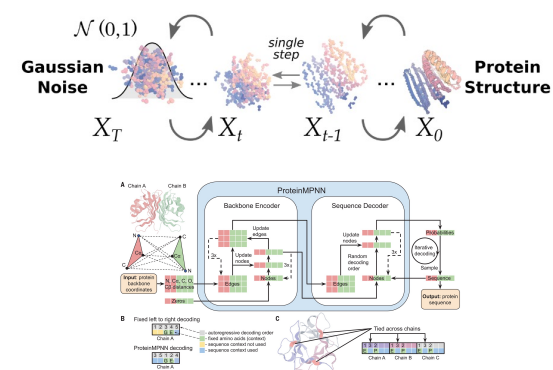
Docking (via LightDock)

Mar 13 (Fri 4-5pm)



PyRosetta tools

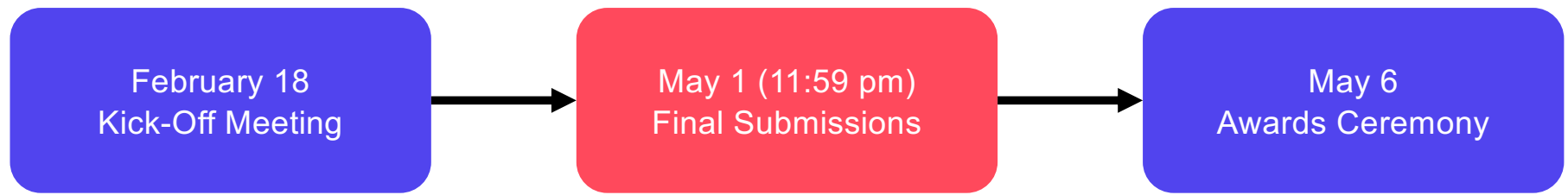
Apr 3 (Fri 4-5pm)



RFdiffusion + ProteinMPNN

Logistics

There are plans to potentially take promising candidates and test them in the lab!



Submission details (only 1 per team please):

- Email to apak@mines.edu with *Protein Design Contest 2026* in the subject
- Create ZIP or TAR with:
 - FASTA Sequence of the binder
 - PDB of the designed binder bound/docked to nitrogenase (PDB 7UT8, Chain B)
 - PDF containing the names of team members (and emails) and a rendered picture of the designed binder bound/docked to nitrogenase

Resources

Primer on proteins: <https://www.youtube.com/watch?v=wwTv8TqWC48>

Another primer on proteins: <https://www.youtube.com/watch?v=PPJ7C3hcnPw>

Protein Data Bank (use as inspiration): <https://www.rcsb.org/>

How to use Alphafold 3: <https://www.youtube.com/watch?v=iGGby6zavAY>

Online converter CIF to PDB: <https://project-gemmi.github.io/wasm/convert/cif2pdb.html>

How to install/use FoldIt Standalone:

https://docs.google.com/document/d/1YLRRmLsxtTkx_THqxb3F5WAZSc-y3Nj72UdJywUzZo4/edit?usp=sharing

How to make pretty renderings in VMD: <https://www.youtube.com/watch?v=5ZiHw4Z5yfY>

(Chimera/pyMOL are both fine for visualization, as well)

Please sign up here to receive slides, recordings, and updates



Good luck and have fun!

Questions?